3.7.1 Internal hardware components of a computer

3.7.1.1 Internal hardware components of a computer

Content	Additional information
Have an understanding and knowledge of the basic internal components of a computer system.	Although exam questions about specific machines will not be asked, it might be useful to base this section on the machines used at the centre.
Understand the role of the following components and how they relate to each other: • processor • main memory • address bus • data bus • control bus • I/O controllers.	
Understand the need for, and means of, communication between components. In particular, understand the concept of a bus and how address, data and control buses are used.	

Content	Additional information
Be able to explain the difference between von Neumann and Harvard architectures and describe where each is typically used.	Embedded systems such as digital signal processing (DSP) systems use Harvard architecture processors extensively. Von Neumann architecture is used extensively in general purpose computing systems.
Understand the concept of addressable memory.	